Recommendation System Project

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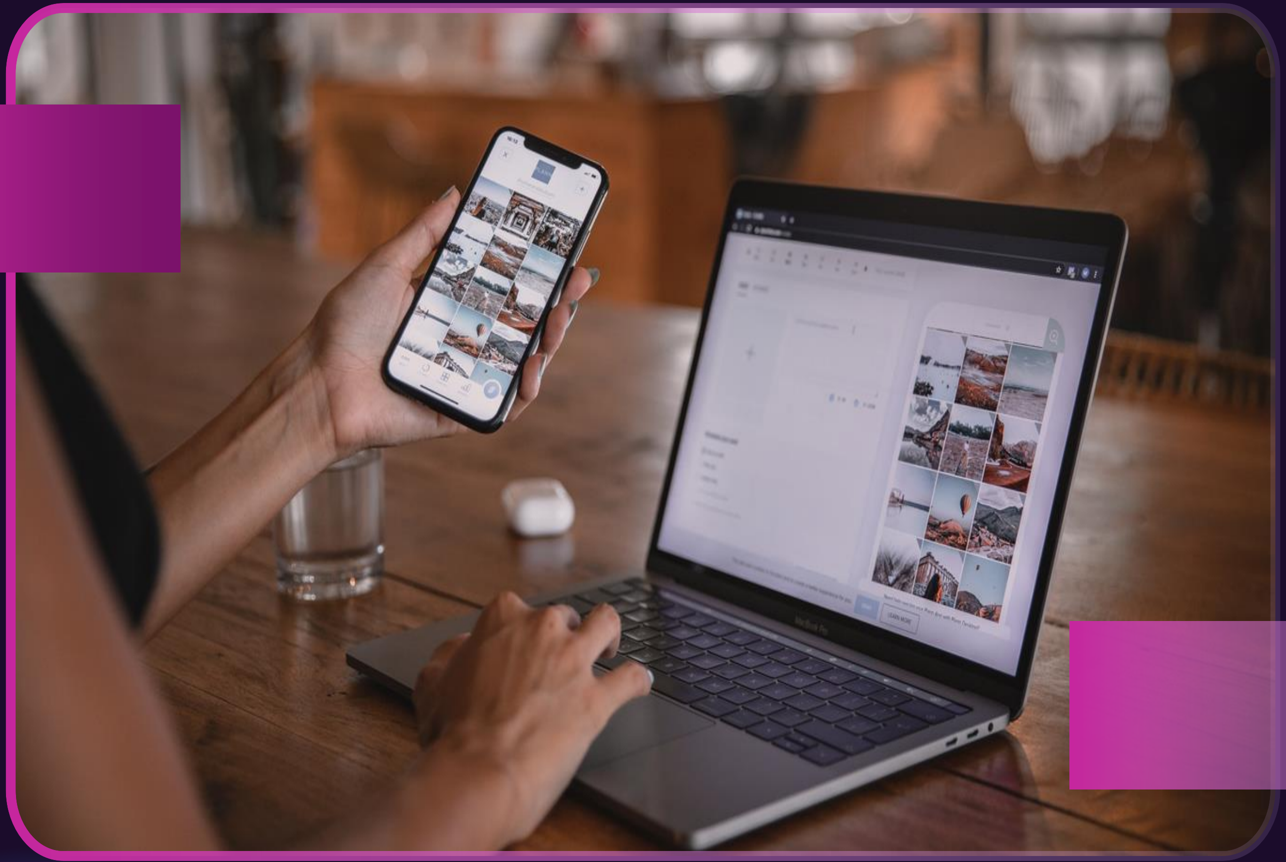
Live Demo

Project Summary



Focus:

- Building a Recommendation System that delivers personalised suggestions based on historical user data, such as past interactions, preferences, and behaviours.

Aim:

- to enhance user satisfaction, engagement, and retention by offering relevant product.

Business Overview & Questions

• Task 1: Recommender System – Hybrid (Rule-based + Logistic Regression)

• Task 2: Anomaly Detection – Isolation Forest

  • Streamlit app with interactive tabs

  • Improved accuracy and fraud detection

01

What categories should we recommend next for a user given their recent behaviour?

02

Which users exhibit abnormal behaviour (e.g., bots/bulk buyers) that should be excluded from modelling and analytics?

03

How much does data cleaning/anomaly removal improve recommendation quality?

Dataset Overview

01

Events Dataset

Timestamp
Visitor_id
Event (view, add to cart, transaction)
Item_id
Transactionid

02

Category Dataset

Categoryid
Parent_id

03

Item Properties 1 & 2

Timestamp
Item_id
Property
Value

04

Sample Dataset

Input columns: timestamp, visitorid, event, categoryid_final
Sample: 30k rows, 200 users

System Flow Diagram

01. Data Ingestion & Cleaning

Load raw events data
Standardise formats
Exploratory data analysis
Removed outliers.

02. Feature Engineering

Construct candidate features
Generate SVD embeddings for users and items to capture latent preferences.

03. Model Training

Train baseline models (Logistic Regression, Random Forest, Gradient Boosting).
Build Hybrid Recommender combining baseline features + SVD embeddings.
Train anomaly detection model to identify abnormal user behaviour.

04. Evaluation

Apply precision, recall, F1 for anomaly detection.
Assess recommender with coverage, top-k hit rate, and accuracy.
Compare "before vs after cleaning" to validate improvements.

05. App Integration(Streamlit)

Develop interactive UI with two key modules:
Recommendations (Top-K personalised categories).
Anomaly Audit (flagging suspicious users).Add debug panels for user activity insights.

06. Deployment

Package artefacts (.pkl, embeddings, metadata, sample data).
Deploy Streamlit app on Streamlit Cloud.

EXPLORATORY DATA ANALYSIS

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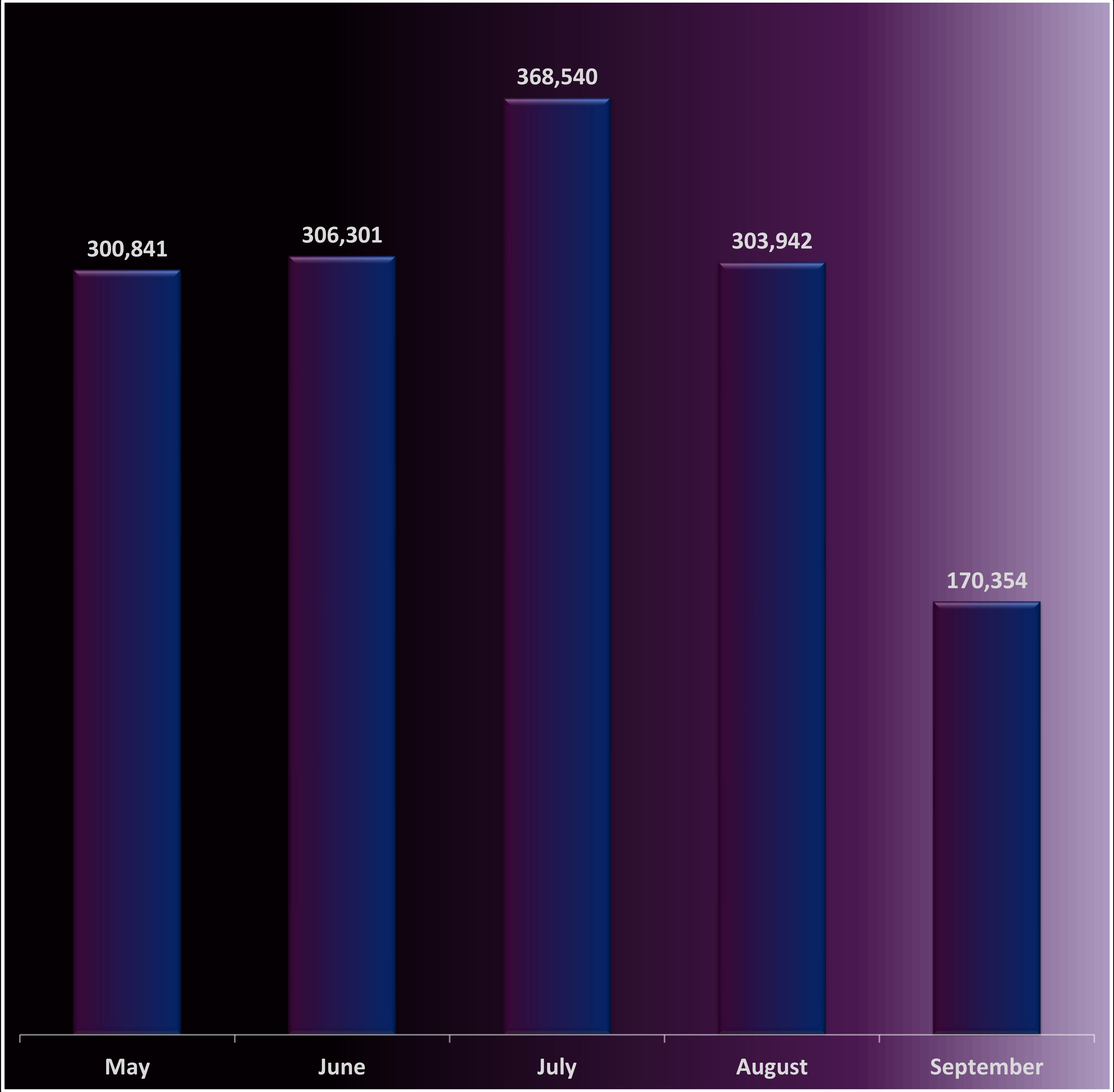

01.

Seasonality and Retention

- User activity peaks in July (368K), while September drops sharply (170K).
- There is a seasonal effect, possibly linked to summer promotions, holidays, or marketing campaigns in July.
- Business could allocate higher budgets to peak seasons and design re-engagement strategies for off-peak months like September.

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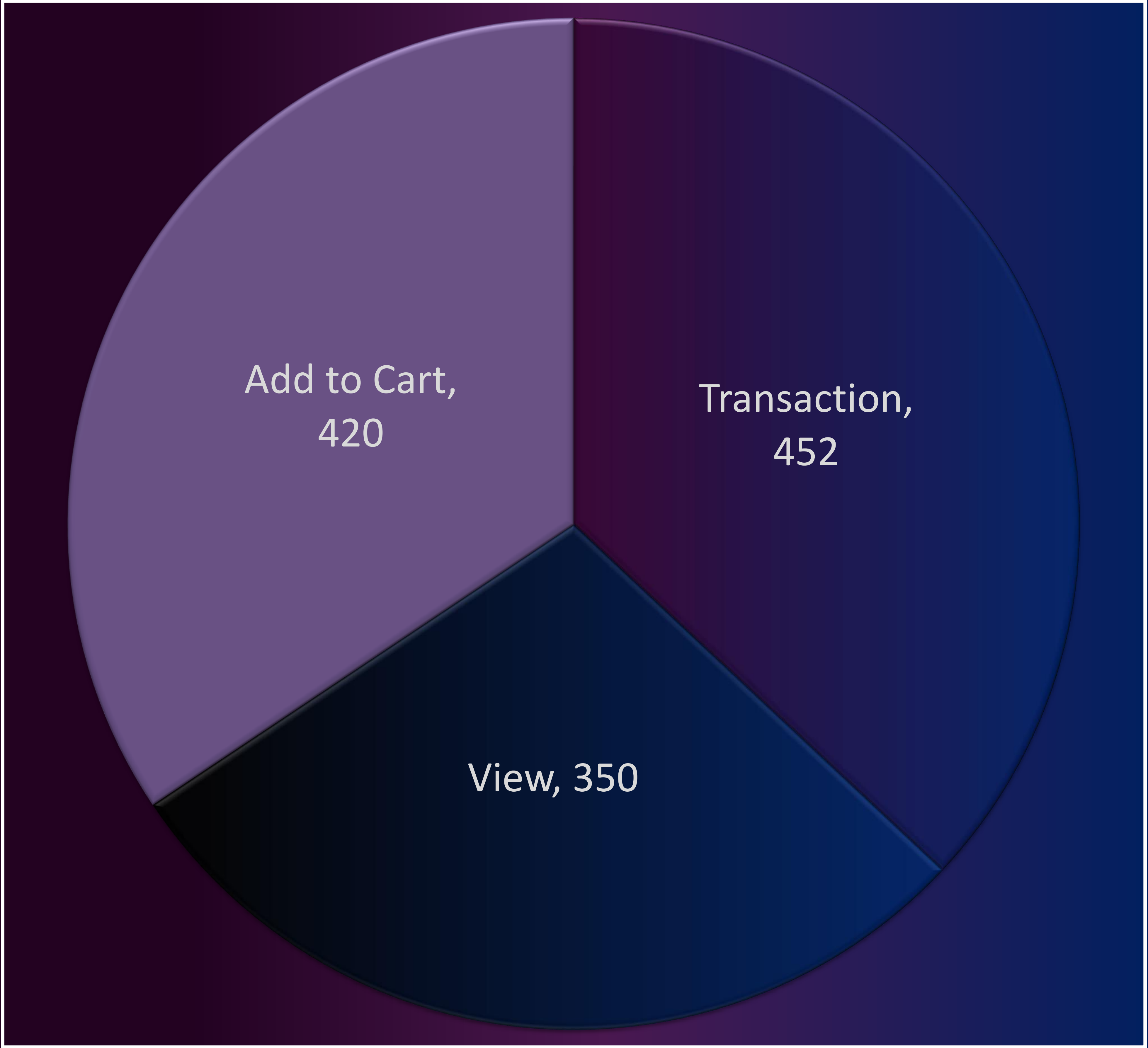
02.

Category Popularity

- Observation: Category 250 dominates with over 6M interactions (606K views alone). Other strong categories include 679, 312, 620, and 561.
- User attention is highly skewed toward a few categories.
- These categories should be prioritised in recommendations, inventory management, and promotions.



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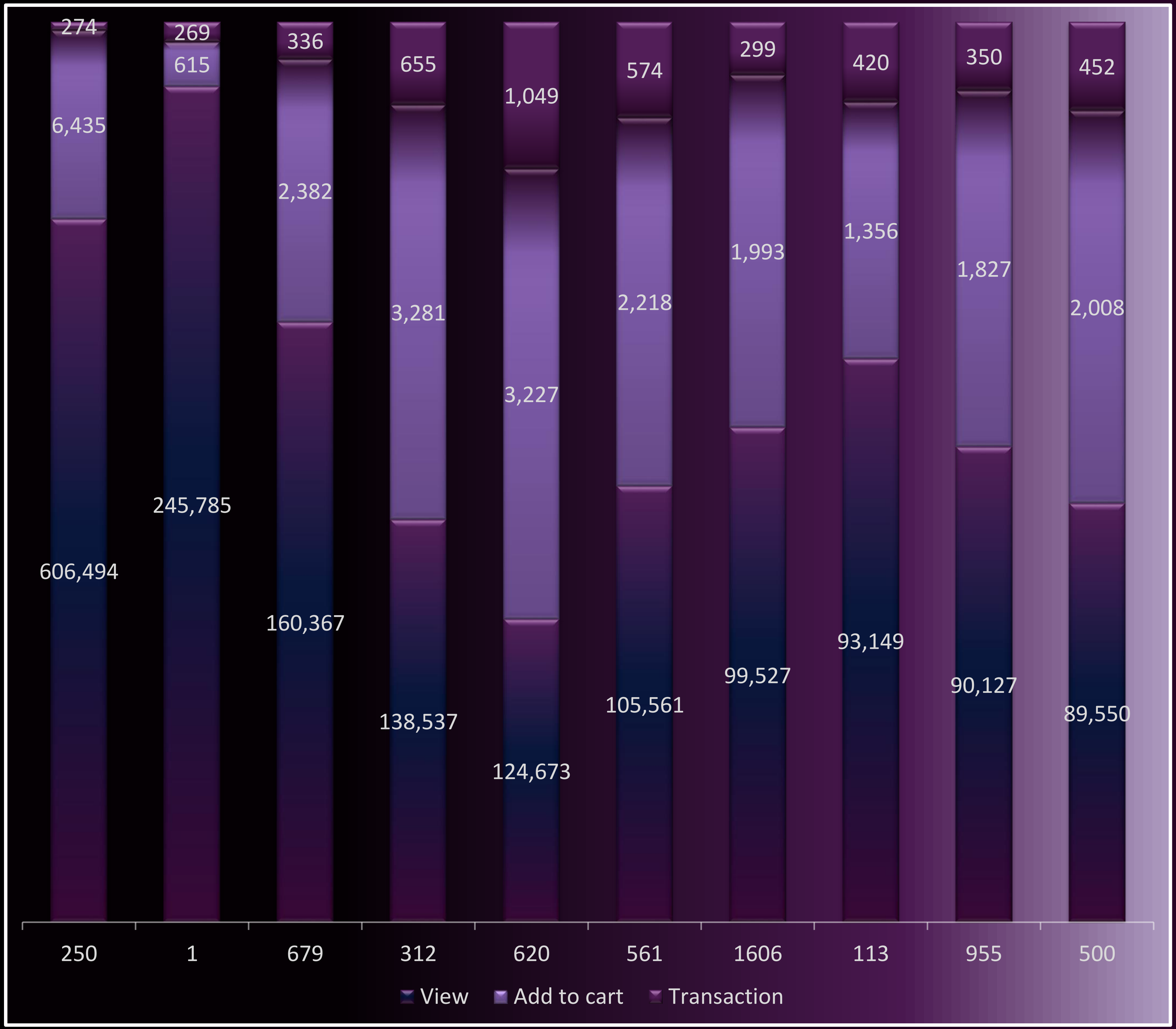

03.

Category Popularity

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MACHINE LEARNING

Feature Engineering

• Preprocessing:
  – Normalized timestamps
  – Dropped invalid rows
  – Sharded embeddings
• Engineered user-level features for anomaly detection

Model Training

Task 1 — Hybrid Recommender:
• Logistic Regression, RandomForest, HistGradient, SVD, Hybrid SVD.

Task 2 — Anomaly Detection:
• Isolation Forest

Task 1 Model Evaluation

- Hybrid SVD (Champion Model)
- Combined handcrafted baseline features + SVD embeddings into a hybrid feature set
- .Performance: Coverage ~91.6%, Top1 ~85.6%, Top3 ~99.5%. Selected as production champion.

Task 2 Model Evaluation

- Proxy anomaly rate of ~2.08%
- Model trained on engineered user behaviour features.
- Chosen contamination: 0.05 → balances recall and precision.
- Performance: Precision = 31%, Recall = 75%, F1 = 0.44.
- Flagged share: 5% of users identified as potentially anomalous.

Live Demo

Streamlit App Deployment

Recommendations

- Upload dataset or use sample
- Pick user, adjust parameters
- Exclude anomalous users

Anomaly Audit

- Load user features
- Run Isolation Forest
- Download flagged users

Business Implications

- Early detection of bots/bulk accounts → protects GMV analytics and inventory signals
- Category recommendations across homepage, product page, and email
- Fraud/abuse mitigation and traffic quality monitoring

Risks & Considerations

- Candidate-label coupling: ensure strict temporal splits & leakage checks.
- Calibrate anomaly threshold to business tolerance; review flagged cohorts regularly.
- Cold-start: add content-based features and back-off rules.
- Monitor drift; schedule retraining (e.g., weekly).

Appendix

Links

- Recommender live Demo Link : https://recommendation-system-project-c8cgulvvsynvm77sfiyfwb.streamlit.app/
- Github Link : https://github.com/julietkukuia/Recommendation-System-Project/tree/main

Thank You!

A recommendation system is only as good as the trust it builds. By combining predictive accuracy with anomaly detection, we not only improve user experience but also safeguard the integrity of the platform.